DASH-03: Executive Dashboards

> **Series:** DASH | **Notebook:** 3 of 7 | **Created:** March 2026 | **Last Updated:** 03/12/2026

Overview

Executive dashboards distill complex observability data into a handful of business-meaningful KPIs. The goal is not to show everything — it is to show exactly what a decision-maker needs to assess system health in under 30 seconds. This notebook covers KPI selection, single-value tiles, trend lines, traffic-light patterns, MTTR calculations, and how to tell a story with data while avoiding information overload.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with Grail enabled
Permissions	`storage:events:read`, `storage:metrics:read`, `storage:spans:read`
Data	Davis problems with closed events (for MTTR), service spans
Prior Reading	DASH-01 and DASH-02

1. KPI Selection Framework

Selecting the right KPIs is the most important decision in executive dashboard design. Use these criteria:

The SMART Filter for Dashboard KPIs

Criterion	Question
Specific	Does this metric answer one clear question?
Measurable	Can Dynatrace calculate it reliably?
Actionable	Does a change in value trigger a specific response?
Relevant	Does the executive audience care about this?
Time-bound	Is the time range meaningful for decision-making?

Recommended Executive KPIs

KPI	Description	Tile Type	Threshold Example
Availability %	Uptime as a percentage of total time	Single value	Green: >99.9%, Yellow: >99.5%, Red: <99.5%
MTTR	Average time to resolve problems	Single value	Green: <1h, Yellow: <4h, Red: >4h
Active Problems	Current open problem count	Single value	Green: 0, Yellow: 1-3, Red: >3
Problem Trend	Problem count over 7-30 days	Line chart	Trending up = concern
Error Rate	Overall % of failed requests	Single value	Green: <1%, Yellow: <5%, Red: >5%

> **Tip:** Start with 4-5 KPIs. You can always add more later, but removing tiles executives have grown accustomed to is harder.

2. Availability Percentage

Availability is the most commonly requested executive metric. There are several ways to calculate it depending on what "availability" means in your organization.

Approach 1: Based on Davis Problem Downtime

Calculate the percentage of time with no AVAILABILITY-category problems.

```
// Availability based on Davis problem downtime over 7 days
fetch dt.davis.problems, from:-7d
| filter event.status == "CLOSED"
| filter event.category == "AVAILABILITY"
| filter dt.davis.is_duplicate == false
| summarize total_downtime_ns = sum(toLong(resolved_problem_duration))
| fieldsAdd total_period_ns = 7.0 * 24 * 3600 * 1000000000
| fieldsAdd availability_pct = round((1 - total_downtime_ns /
total_period_ns) * 100, decimals: 3)
```

Approach 2: Based on Service Success Rate

Calculate availability as the percentage of successful server-side requests.

```
// Service-level availability based on span success rate
fetch spans, from:-24h
| filter span.kind == "server"
| summarize total = count(), successes = countIf(otel.status_code != "ERROR")
| fieldsAdd availability_pct = round(100.0 * successes / total, decimals: 3)
```

3. Mean Time to Resolve (MTTR)

MTTR measures how quickly your team resolves problems. It is one of the four DORA-adjacent reliability metrics that executives track.

Current MTTR (Single Value)

```
// Average MTTR over last 7 days in hours
fetch dt.davis.problems, from:-7d
| filter event.status == "CLOSED"
| filter dt.davis.is_frequent_event == false
| filter dt.davis.is_duplicate == false
| summarize mttr_hours = avg(toLong(resolved_problem_duration) /
3600000000000.0)
```

MTTR Trend Over Time

Show MTTR as a daily trend to highlight improvement or degradation.

```
// MTTR trend — daily average over 30 days
fetch dt.davis.problems, from:-30d
| filter event.status == "CLOSED"
| filter dt.davis.is_frequent_event == false AND dt.davis.is_duplicate ==
false
| makeTimeseries mttr_hours = avg(toLong(resolved_problem_duration) /
3600000000000.0), interval:1d, time:event.end
```

4. Problem Count and Trends

Problem trends reveal whether operational health is improving or declining over time.

Weekly Problem Trend by Category

```
// Problem trend by category over 7 days
fetch dt.davis.problems, from:-7d
| filter dt.davis.is_duplicate == false
| makeTimeseries problem_count = count(), interval:1d, by:{event.category}
```

Problems by Severity — Executive Summary Table

```
// Problem summary by category — executive table tile
fetch dt.davis.problems, from:-7d
| filter dt.davis.is_duplicate == false
| summarize total = count(), active = countIf(event.status == "ACTIVE"),
closed = countIf(event.status == "CLOSED"), by:{event.category}
| sort total desc
```

5. Service Health Score

A composite health score combines multiple signals into a single number. This is useful for executives who want one metric per service.

Simple Health Score Formula

Health = 100 - (error_rate_penalty + latency_penalty)

Factor	Penalty Calculation
Error rate > 1%	+20 penalty
Error rate > 5%	+50 penalty
P95 latency > 1s	+15 penalty
P95 latency > 3s	+35 penalty

```
// Service health score — composite metric for executive dashboard
fetch spans, from:-1h
| filter span.kind == "server"
| summarize total = count(), errors = countIf(otel.status_code == "ERROR"),
p95_ns = percentile(duration, 95), by:{dt.entity.service}
| fieldsAdd error_rate = 100.0 * errors / total
| fieldsAdd p95_ms = p95_ns / 1000000
| fieldsAdd error_penalty = if(error_rate &gt; 5, then: 50, else:
if(error_rate &gt; 1, then: 20, else: 0))
| fieldsAdd latency_penalty = if(p95_ms &gt; 3000, then: 35, else: if(p95_ms
&gt; 1000, then: 15, else: 0))
| fieldsAdd health_score = 100 - error_penalty - latency_penalty
| fieldsKeep dt.entity.service, health_score, error_rate, p95_ms
| sort health_score asc
| limit 10
```

6. Error Budget Tracking

Error budgets quantify how much failure is acceptable before an SLA breach. For a 99.9% SLA, the monthly error budget is 43.2 minutes of downtime.

SLA Target	Monthly Error Budget
99.99%	4.32 minutes
99.9%	43.2 minutes
99.5%	3.6 hours
99.0%	7.2 hours

Error Budget Remaining

```
// Error budget remaining for 99.9% SLA target — 30-day window
fetch dt.davis.problems, from:-30d
| filter event.status == "CLOSED"
| filter event.category == "AVAILABILITY"
| filter dt.davis.is_duplicate == false
| summarize total_downtime_min = sum(toLong(resolved_problem_duration)) /
60000000000.0
| fieldsAdd budget_total_min = 43.2
| fieldsAdd budget_remaining_min = budget_total_min - total_downtime_min
| fieldsAdd budget_remaining_pct = round(100.0 * budget_remaining_min /
budget_total_min, decimals: 1)
```

7. Storytelling with Data

An executive dashboard should tell a story without explanation. Follow these storytelling principles:

Layout Pattern for Executive Dashboard

Row	Left	Center	Right
Top	Availability % (single value)	MTTR (single value)	Active Problems (single value)
Middle	Problem trend (7d line chart)	—	Error budget remaining (single value)
Bottom	Service health table (top 5 worst)	—	—

Storytelling Principles

1. **Lead with the headline** — the top row answers "are we healthy?" in 3 numbers
2. **Support with trends** — the middle row shows direction (improving or declining)
3. **Provide detail on demand** — the bottom row offers specifics for those who want them

4. **Use color intentionally** — green means healthy, yellow means watch, red means act
5. **Avoid decoration** — no gratuitous pie charts, no unnecessary 3D effects, no logos

> **Warning:** Executives will lose trust in a dashboard that shows misleading data. Always validate that your availability and MTTR calculations match what the team reports manually. A discrepancy between the dashboard and a status meeting destroys credibility.

8. Summary and Next Steps

In this notebook you learned:

- How to select KPIs using the SMART filter
- Two approaches to calculating availability percentage
- MTTR calculation and trending over time
- Problem count analysis by category and status
- Composite service health scores
- Error budget tracking for SLA targets
- Storytelling principles for executive dashboard layout

Next: In **DASH-04: Operations Dashboards**, we move to the operations tier — real-time service monitoring, infrastructure health, log volume tracking, and problem/alert integration.
